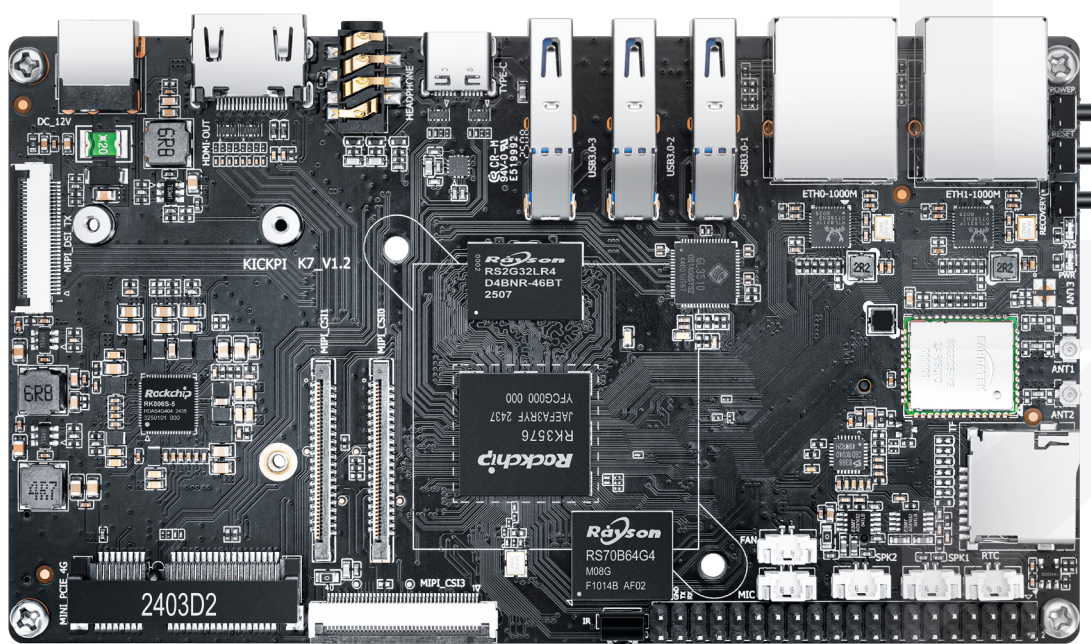


KICKPI

KICKPI-K7

RK3576 Product Specification



Shenzhen Kickpi Digital Technology Co., Ltd

Catalogue

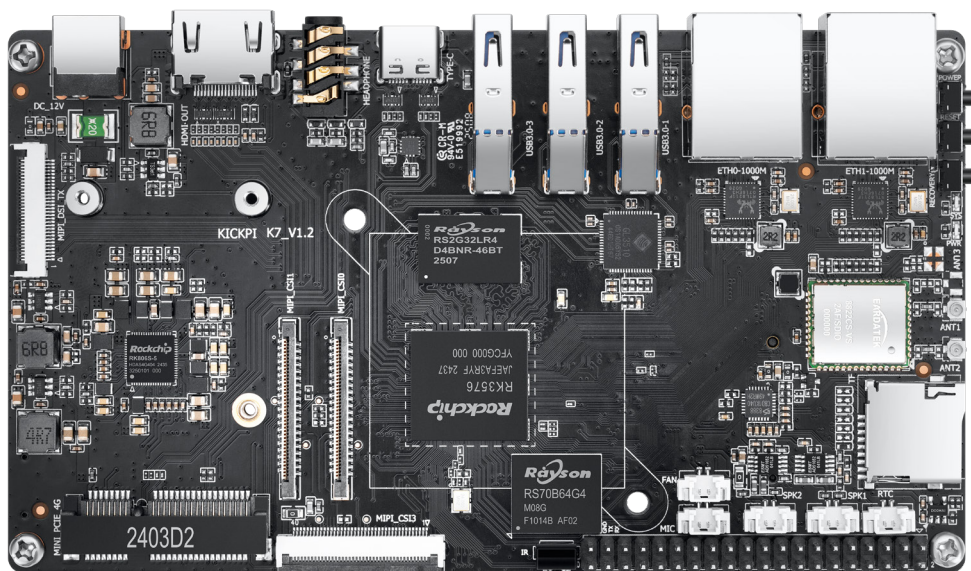


KICKPI

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Product Features

01



Strong Core Power



- Cortex-A72 X4 + Cortex-A53 X4
- Mali-G52 GPU, enjoy smooth graphics processing
- Supported by 6TOPS NPU, it offers powerful computing power

High-speed Interconnection



- 4G/5G full network compatibility, smooth connection to mobile networks anytime and anywhere
- Dual Gigabit Ethernet, wired dual channels, stable and low latency
- Dual-band WIFI Bluetooth, 2.4G/5G transmission is smoother

Multi-screen Intelligent Control, free Expansion



- HDMI 4K@120Hz, the first choice for ultra HD large screens
- TYPE-C DP 4K@120Hz, one-line direct connection, efficient and convenient
- MIPI-DSI 1080P@60Hz, compatible with mobile devices

Multi-system Compatibility Worry-free Development



- Android 14 has a rich mobile application ecosystem and optimized touch interaction
- Debian 12 is stably open source and factory compatible
- Ubuntu 24.04 is developer friendly

Efficient Development Deep Customization



- The underlying driver is open, with complete source code and technical documentation
- Expert-level technical support, with senior engineers providing one-on-one coordination
- Double the development efficiency, provide standard BSP package

Full-scenario Empowerment Industry Customization Experts

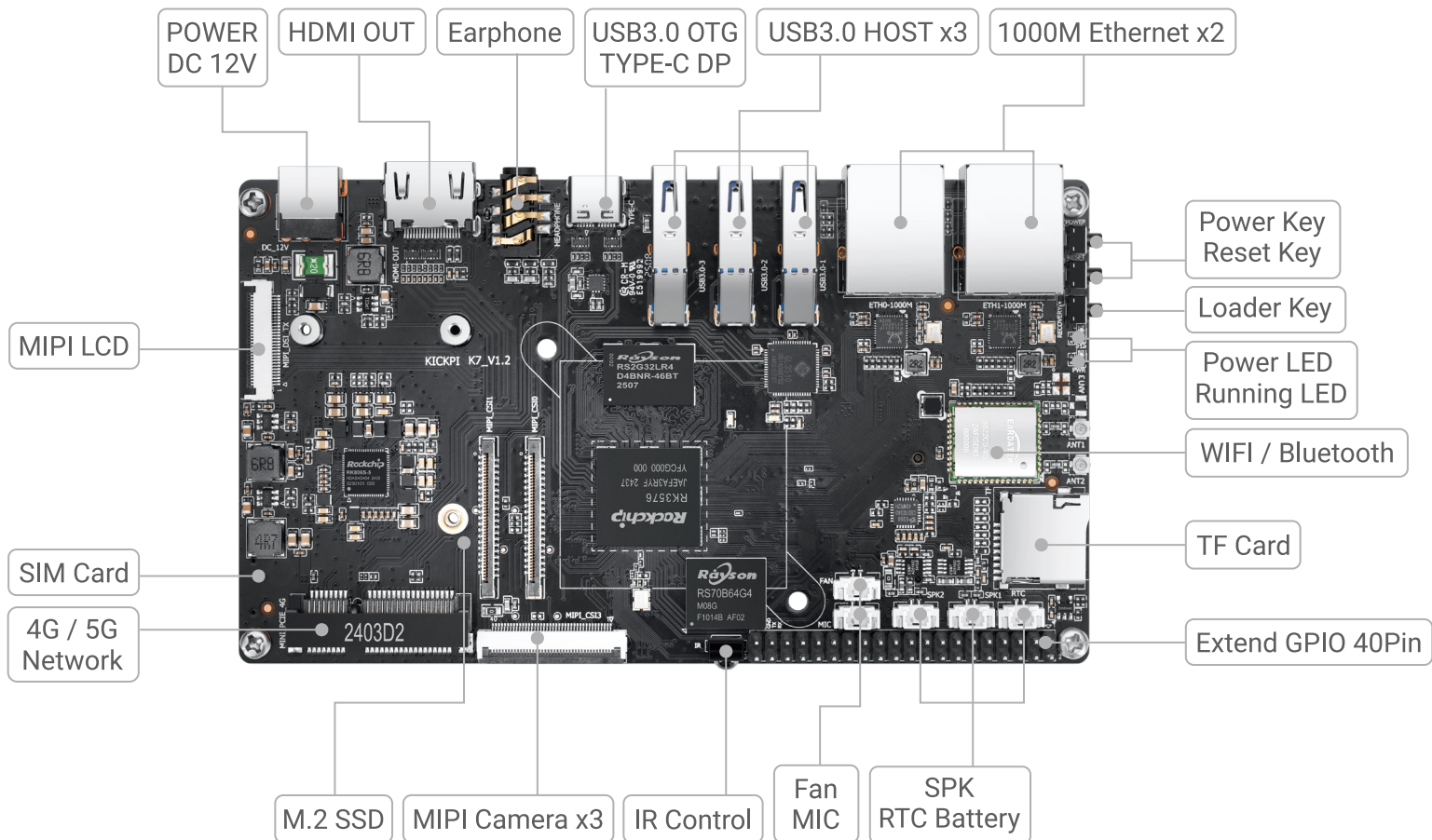


- Smart Internet of Things: Cost-effective gateways, edge computing nodes
- Industry 4.0: Industrial control tablets, industrial inspection equipment
- Business Entertainment: portable TV, tablets

01

Interface Description

02



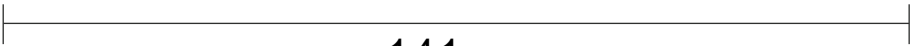
Specifications

03

Type	Main Specifications
Main Controller	Rockchip RK3576
CPU	Quad Cortex-A72 and quad Cortex-A53
GPU	ARM Mali G52 MC3
NPU	RKNN 6 Tops
Memory	4GB / 8GB / 16GB
Storage	16GB / 32GB
Systems Supported	Android 14.0 / Debian 12 / Ubuntu 24.04
Power Supply	DC 12V
Cooling Fan	Y
LED Light	* Power LED * Work LED
USB HOST	USB 3.0 HOST x3
USB TYPE-C	USB3.1 OTG / USB Debug / DP v1.4
Infraed Receiver	Y
Real-Time Clock	Y (HYM8563)
Debug Serial Port	Y
Buttons	* Recovery KEY * RESET KEY * MASKROM KEY * POWER KEY
Expansion Pins	Extend Pin Total 40Pin: *UART x5 *SPI x1 *PWM x7 *ADC x3 *I2C x3 *I3C x1 *GPIO x22 *PDM x1 *CAN x2
HDMI Display	x1 (up to 4K@120Hz)
MIPI DSI Display	x1 (up to 1920*1080@60Hz)
MIPI CSI Camera	4 Lane x4
SPK	Y
Earphone	Y
MIC	Y
WIFI/Bluetooth	Y (RTL8822CS)
Ethernet	1000M x2
Mobile Network	Y (4G EC20 / EC200) (5G RG200U)
TF card	Y
M.2 Hard Drive	Y

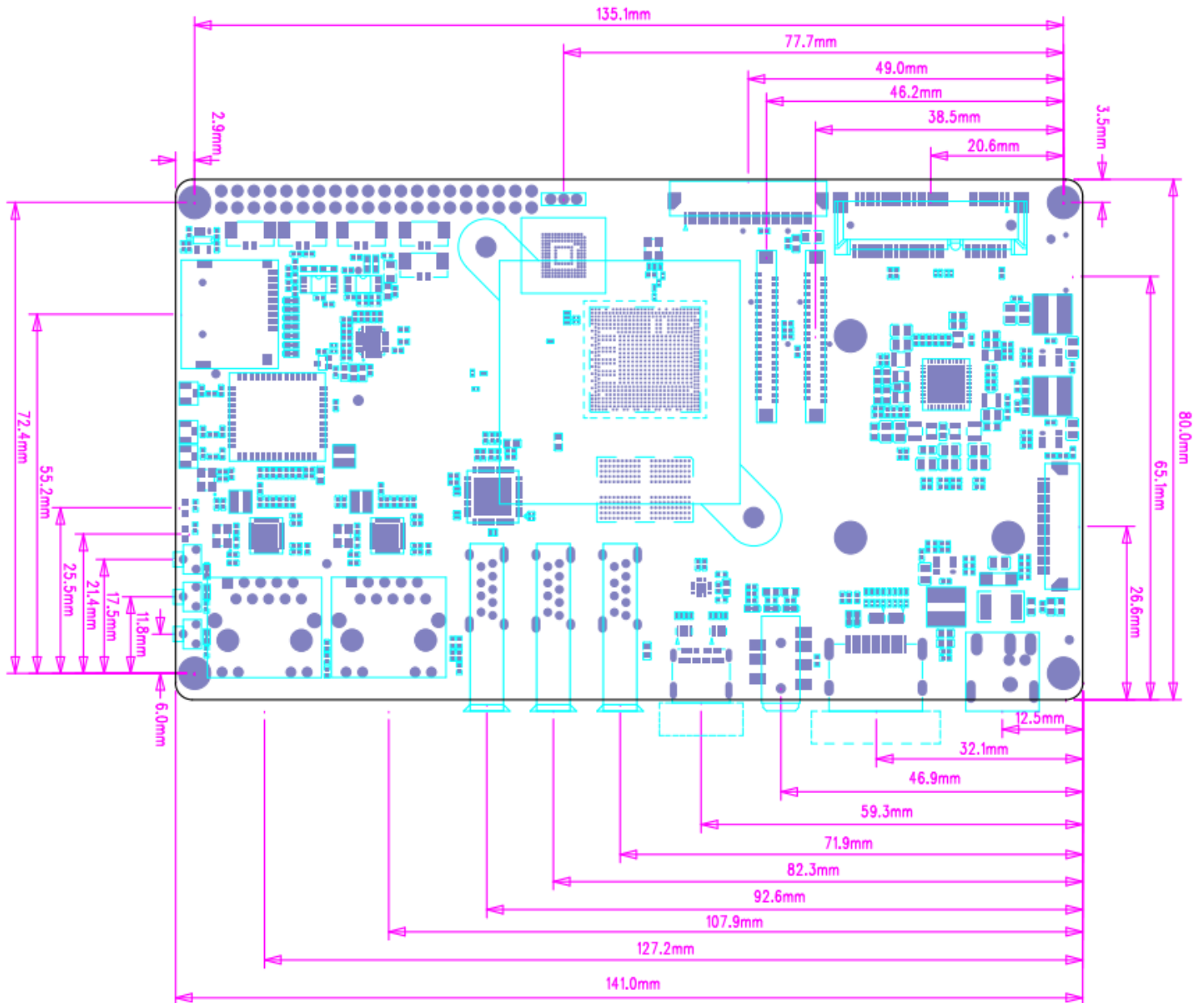
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C



PCB Dimension

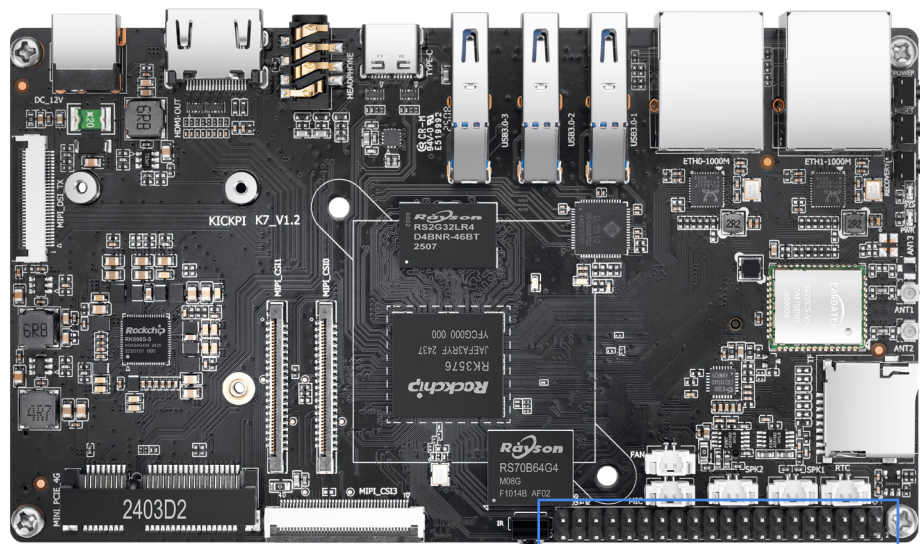
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05

Interface Illustration

06



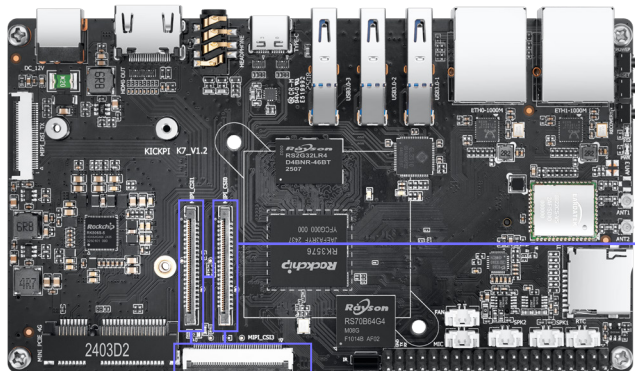
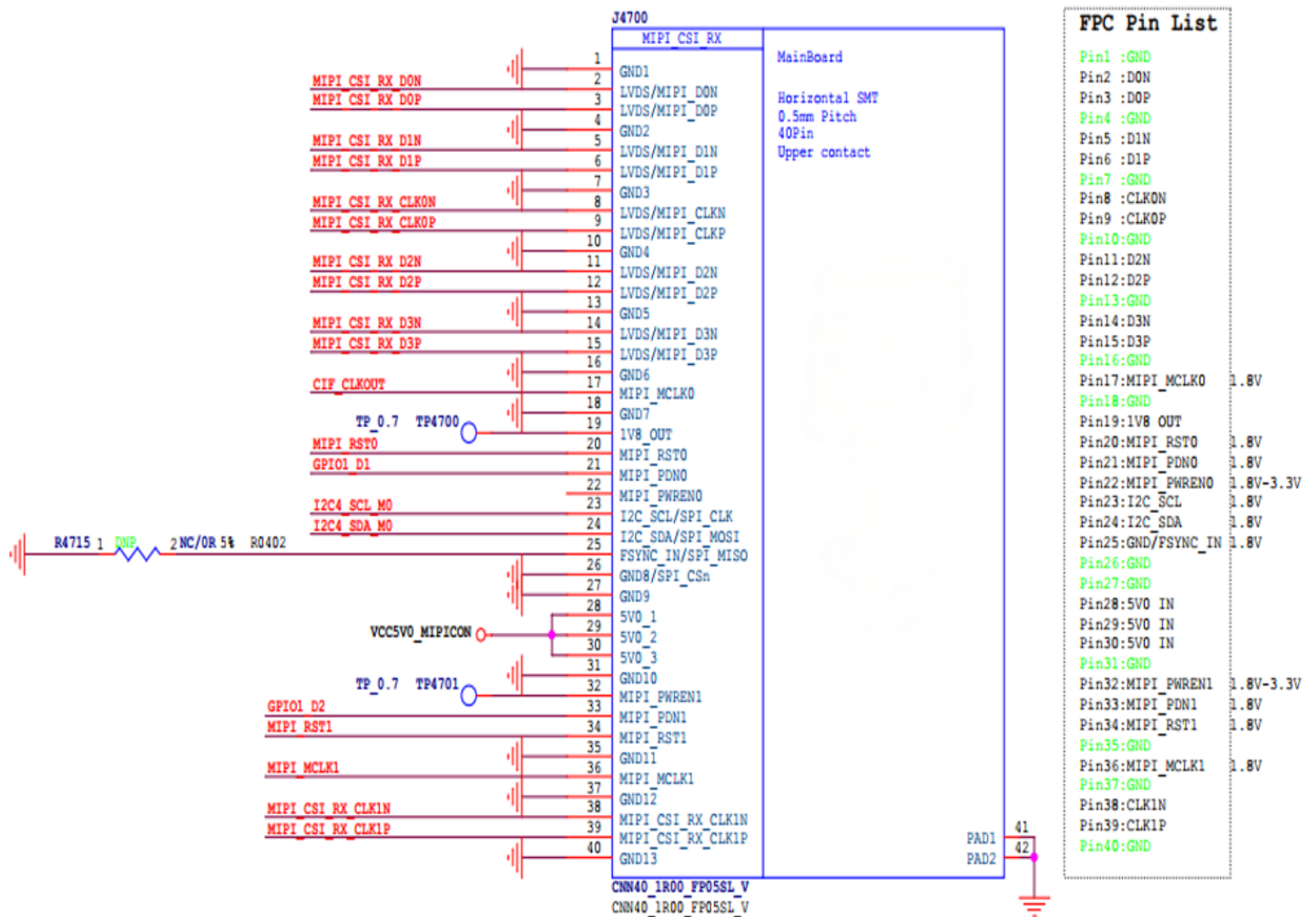
40Pin Definition

		VCC_3V3	1		2	VCC_5V0		
		VCC_3V3	3		4	VCC_5V0		
UART6_TX_M0	CAN0_TX_M2	GPIO4_A4	5		6	GND		
UART6_RX_M0	CAN0_RX_M2	GPIO4_A6	7		8	GPIO4_B0	PDM1_CLK1_M1	*SPI4_CLK_M2
	*PWM2_CH7_M2	GPIO2_D7	9		10	GPIO4_B1	PDM1_SDI2_M1	*SPI4_MOSI_M2
	*PWM2_CH6_M2	GPIO2_D6	11		12	GPIO4_B2	PDM1_SDI1_M1	*SPI4_MISO_M2
		GND	13		14	GPIO0_C3	*PWM0_CH1_M0	
		VCC_3V3	15		16	GPIO1_D0	UART10_TX_M1	*SPI_CS
PWM2_CH2_M3	*UART8_RX_M0	GPIO3_C5	17		18	GPIO1_D1	UART10_RX_M1	*I3C0_SDA_PU_M1
	*UART8_TX_M0	GPIO3_C6	19		20	GND		
I2C3_SDA_M2	UART5_TX_M0	*GPIO3_D5	21		22	GPIO3_A0	*I2C7_SCL_M1	UART3_TX_M0
I2C3_SCL_M2	UART5_RX_M0	*GPIO3_D4	23		24	GPIO3_A1	*I2C7_SDA_M1	UART3_RX_M0
		VCC_3V3	25		26	GPIO1_D2	*I3C0_SCL_M1	PWM1_CH3_M1
		GND	27		28	GPIO1_D3	*I3C0_SDA_M1	PWM1_CH4_M1
		*GPIO0_A5	29		30	*GPIO3_DO	PWM1_CH0_M3	
		GND	31		32	*GPIO3_A4	PWM2_CH3_M3	
		*GPIO3_B0	33		34	GND		
		DEBUG_UART0_RX	35		36	*SARADC_VIN4		
		DEBUG_UART0_TX	37		38	*SARADC_VIN5		
		GND	39		40	*SARADC_VIN6		

* Marked as the default pin multiplexing function

06

MIPI CSI Camera Interface Definition

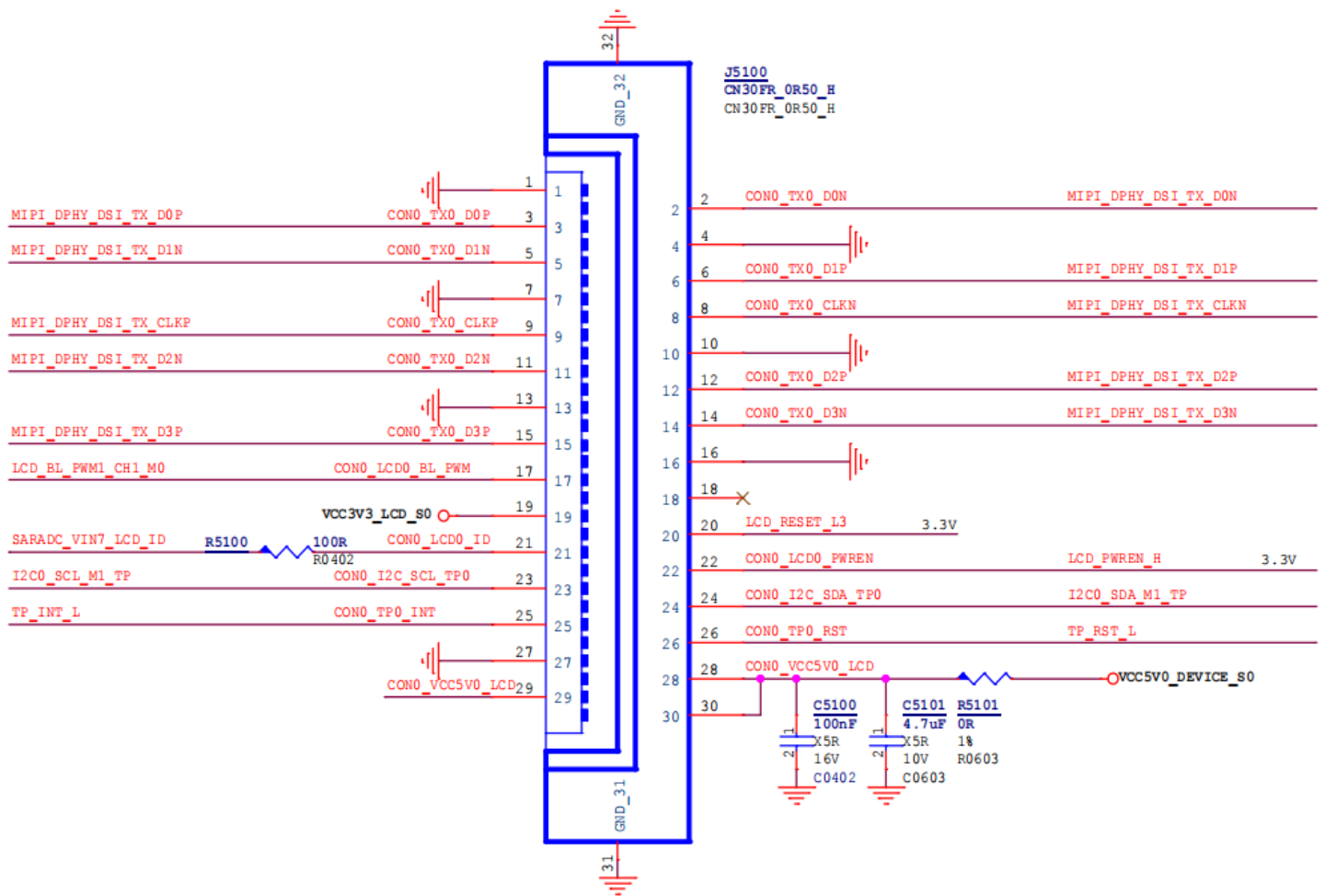


- The development board is equipped with three MIPI CSI 4 Lane camera interfaces.
- The RK3576 has a built-in high-performance ISP processor.
- Supports up to 16 megapixels.

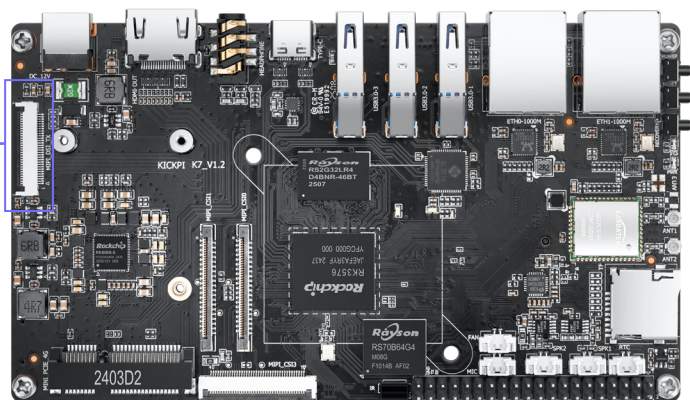
Pin Definition

1	GND	21	MIPI_PDN0
2	D0N	22	MIPI_PWREN0
3	D0P	23	I2C_SCL
4	GND	24	I2C_SDA
5	D1N	25	GND / FSYNC_IN
6	D1P	26	GND
7	GND	27	GND
8	CLKON	28	5V0 IN
9	CLKOP	29	5V0 IN
10	GND	30	5V0 IN
11	D2N	31	GND
12	D2P	32	MIPI_PWREN1
13	GND	33	MIPI_PDN1
14	D3N	34	MIPI_RST1
15	D3P	35	GND
16	GND	36	MIPI_MCLK1
17	MIPI_MCLK0	37	GND
18	GND	38	CLK1N
19	1V8_OUT	39	CLK1P
20	MIPI_RST0	40	GND

MIPI/LVDS LCD Interface Definition



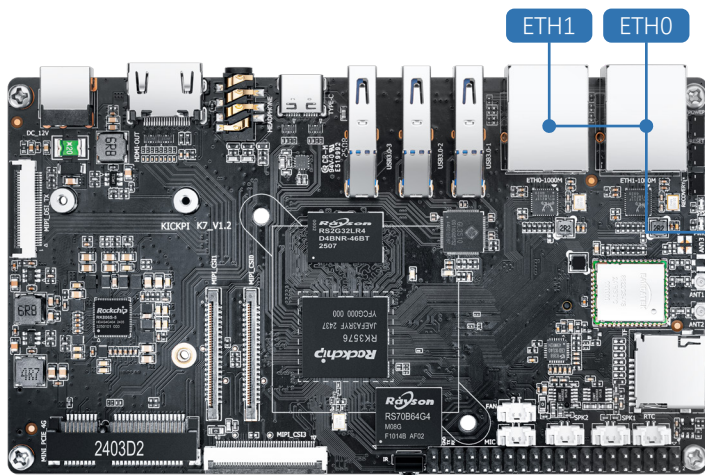
- The development board supports the MIPI LCD interface.
- The highest resolution of the MIPI LCD is 1920x1080@60Hz.



Pin Definition

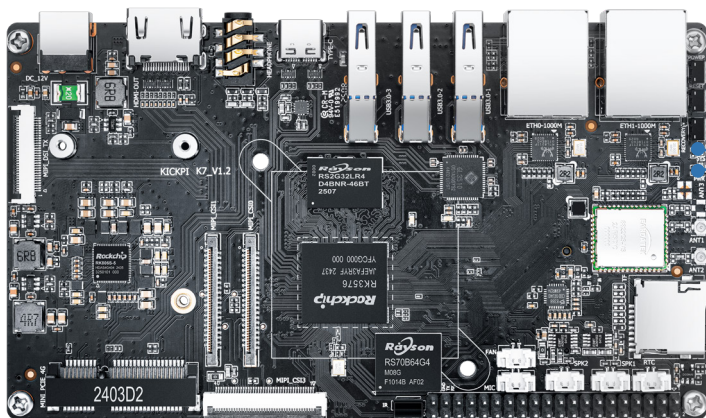
1	GND	16	GND
2	D0N	17	LCD_PWM_BL
3	D0P	18	LCD_TE
4	GND	19	VCC3V3_LCD
5	D1N	20	LCD_RST
6	D1P	21	LCD_ID
7	GND	22	LCD_PWREN
8	CLKN / AUXN	23	TP_I2C_SCL
9	CLKP / AUXP	24	TP_I2C_SDA
10	GND	25	TP_INT
11	D2N	26	TP_RST
12	D2P	27	GND
13	GND	28	5V0
14	D3N	29	5V0
15	D3P	30	5V0

Ethernet



The development board is equipped with two Gigabit Ethernet interfaces

LED



- The development board is equipped with two LED indicators, which can easily check the operating status of the development board.
- The green LED is the power indicator, and the connected power adapter is steady on by default.
- The blue LED is the heartbeat light of the system. When the core is running normally, it keeps blinking.

Important Notes

07

Operating Ambient

Operation Temperature	0~80℃
Operating Voltage	12V
Power Dissipation(Bare board)	1.3W

Notes

1. Please be careful not to operate the wiring with electricity during the assembly process.
2. When coming into contact with the PCBA motherboard, it is necessary to wear static protection tools such as an electrostatic wristband (sleeve).
3. When connecting external devices to the motherboard, it is necessary to strictly check the PIN definition and avoid any errors or reverse connections.
4. During the installation and fixation process, avoid structural issues that may cause deformation of the board.
5. When installing the LCD screen, be sure to pay attention to the selection of screen voltage and current, and pay attention to the position of the first pin.
6. When installing the LCD screen, be sure to pay attention to the backlight voltage and whether the power supply is within the power range of the power adapter.
7. When connecting external devices, pay attention to the level matching of external device data, the size of current, and other issues.
8. When installing the serial port, pay attention to what type of serial port device is connected and whether the pins of TX and RX are reversed.
9. Considering the overall power, what is the overall power and whether the power supply is sufficient after the entire device is connected.

Contact Us

08



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